# RARE OBSERVATIONS UNDER LATTICE-SUM CONDITIONING AND JOINT ALLOCATION EXTREMES

YITZCHAK SHMALO

Abstract. For independent lattice-valued observations with finite positive variance, conditioning the sum in a bounded central-limit window preserves the complete indexed record of any event whose probability tends to zero, in total variation. We prove a triangular-array version and a joint extension with an additional $o(n)$-coordinate sample under finitely many lattice constraints. Consequences include approximation of every $o(n)$ upper order statistics, Poisson approximation of all rare-value counts even when their total mean diverges, and ranked outdegree comparisons for Galton–Watson trees conditioned on size and a central leaf count. The allocation results answer Janson's fixed-rank question, including its moving-tilt and weak-centering formulations. Counterexamples show that a positive fraction of ranks can retain the sum constraint and that the fixed-tilt logarithmic centering scale cannot generally be enlarged.

## 1. Introduction

Conditioning a sample on its sum creates dependence between all its coordinates. Nevertheless, its largest observations may have almost the same joint law as before conditioning. We prove this by comparing the complete record of observations in a rare set, including their positions. The comparison holds in total variation, uniformly over attainable sums in a bounded central-limit window, and does not require the expected number of recorded observations to stay bounded.

For probability measures on a countable space, write

$$d_{\mathrm{TV}}(P,Q) = \sup_A |P(A) - Q(A)| = \frac{1}{2}\sum_x |P\{x\} - Q\{x\}|. \tag{1.1}$$

Order statistics are arranged in decreasing order, with multiplicity. For a lattice law, its *maximal span* is the largest $d > 0$ for which its support lies in some translate $a + d\mathbb{Z}$.

**Theorem 1.1** (Sublinear collections of upper ranks)**.** *Let $X_1, X_2, \dots$ be independent copies of a nonnegative integer-valued random variable $X$ with mean $\mu$ and variance $0 < \sigma^2 < \infty$. Put $S_n = \sum_{i=1}^n X_i$. For every fixed $C > 0$ and every integer sequence $1 \le k_n = o(n)$,*

$$\sup_{\substack{m:\ \mathbb{P}(S_n=m)>0 \\ |m-n\mu|\le C\sqrt{n}}} d_{\mathrm{TV}}\Big(\mathcal{L}((X_{(j)})_{j\le k_n} \mid S_n = m), \mathcal{L}((X_{(j)})_{j\le k_n})\Big) \longrightarrow 0. \tag{1.2}$$

No regular variation, smoothness of the mass function, or moment of order greater than two is assumed. Gaps in the support and ties between order statistics are permitted. For a fixed bounded law, both rank vectors equal its maximum support value with probability tending to one. For unbounded laws, the conclusion compares the complete joint distribution of a growing collection of nonconstant order statistics. For example, $k_n = \lfloor n^{0.9} \rfloor$ and $k_n = \lfloor n/\log n \rfloor$ are allowed. These are asymptotic statements, not finite-sample error bounds.

### 1.1. The allocation question.

Whenever the generating function admits a positive exponential tilt, a weighted allocation is an independent sample from the tilted law, conditioned on its total. Janson [9, Theorem 19.7] proves total-variation approximation for the largest box and asks whether it extends to every fixed upper rank; the surrounding discussion also proposes the joint extension to finitely many ranks. This is Problem 19.10 in the published survey, numbered 18.10

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in its 2011 preprint. Theorem 1.1 gives the fixed-law conclusion and extends the number of ranks from fixed to arbitrary $o(n)$. Section 6 treats mean-matched moving tilts and the finite-variance boundary. Section 7 proves the weaker interior fixed-tilt centering statement. The adjacent sparse-allocation question and infinite-variance regimes are outside the present results.

1.2. **Related conditional approximations.** Conditional limit theorems have a substantial literature. Holst [7] studies conditional limits with applications to occupancy-type problems. Diaconis and Freedman [5] give total-variation approximations for predetermined coordinate blocks in exponential families. Their hypotheses include regularity and moment assumptions suited to quantitative estimates. Our observation instead combines a predetermined block with coordinates selected through their values; the fixed-law theorem requires only a second moment and a lattice nondegeneracy condition.

Arratia and Tavaré [3, Sections 2–3 and 8] develop a general framework for independent processes conditioned on weighted sums, including vector constraints and further conditioning. In particular, their Theorem 3 expresses total variation through a ratio of residual-sum probabilities. The conditioning representation and likelihood-ratio method used here belong to that framework; they are not new principles. Our contribution is the uniform finite-variance comparison for a value-selected indexed record, its joint version with a growing coordinate block, and the resulting range of upper ranks.

Asymptotic independence between sums and extreme observations also predates the allocation question. Hsing [8] treats sums and rare values of weakly dependent stationary sequences. Li and Tan [11] study joint limit theorems for extreme order statistics and partial sums of independent observations. Such joint-limit results motivate the present problem, but weak convergence alone does not control conditioning on one lattice value, whose probability tends to zero.

The closest allocation comparison is Stufler's result for a growing collection of extreme component sizes in Gibbs partitions [13, Corollary 3.6]. The number of components there is random, and Remark 3.9 explicitly relates conditioning on that number to Janson's question. Averaging over the number of summands does not by itself give a comparison at each prescribed number. We keep that number fixed and obtain every $o(n)$ upper ranks under finite variance. The admissible rank range and model assumptions should therefore be compared together, rather than by the phrase "growing extremes" alone.

For conditioned trees, the allocation representation comes from the classical cyclic argument underlying the total-progeny formula [6, 9]. Thévenin [14] studies vertices with specified outdegrees and related conditioned-tree limits. Kortchemski and Vetter [10] study rare fringe-subtree patterns, using local-limit estimates to remove bridge conditioning. Their patterns involve several adjacent vertices, unlike the single-coordinate record here. At the opposite, large-deviation scale, Armendáriz and Loulakis [1] describe conditioned heavy-tailed samples with a large summand. Those condensation results concern a different conditioning regime from the central windows considered below.

The proofs combine this conditioning approach with a uniform lattice local limit theorem and Scheffé's argument [12]. The Poisson corollary uses the classical binomial–Poisson bound [4, 2].

1.3. **Proof mechanism.** After the rare values and their positions are revealed, the unrevealed variables remain independent with the law conditioned to avoid the rare set. Bayes' formula therefore expresses the conditional record density as a ratio of two lattice-sum probabilities. The centered sum of the removed observations fluctuates on an $o(\sqrt{n})$ scale; its expectation need not be negligible. An exact centering identity makes the two local Gaussian displacements agree uniformly. A one-sided identity for normalized densities then converts convergence in probability of the likelihood ratio into total-variation convergence. Sorting a sufficiently rich rare record gives the upper-rank theorem.

The general record theorem is stated in Section 2 and proved in Section 4, using the local limit theorem of Section 3. Sections 5–7 give the allocation applications. Section 8 gives Poisson approximations. Section 9 treats several constraints and a growing coordinate block, with an application to trees with a prescribed leaf count. Section 10 establishes the limits of the conclusions. The finite checks in Appendix A supplement the analytic proofs.

## 2. The rare-record theorem

For each $n$, let $X_{n,1}, \dots, X_{n,n}$ be independent with common law $P_n$, mean $\mu_n$ and variance $\sigma_n^2$. We impose the following assumptions:

(i) For fixed integers $a$ and $d \geq 1$, every $P_n$ is supported on $a + d\mathbb{Z}$.

(ii) $X_{n,1} \Rightarrow X$, where $X$ is nondegenerate and has maximal span $d$.

(iii) The squares are uniformly integrable:

$$\lim_{M\to\infty} \sup_n \mathbb{E}[X_{n,1}^2 \mathbf{1}_{\{|X_{n,1}|>M\}}] = 0. \tag{2.1}$$

These assumptions imply $\mu_n \to \mu := \mathbb{E}X$ and $\sigma_n^2 \to \sigma^2 := \operatorname{Var} X \in (0, \infty)$.

Let $A_n \subset a + d\mathbb{Z}$ be any deterministic sets with

$$q_n := \mathbb{P}(X_{n,1} \in A_n) \longrightarrow 0. \tag{2.2}$$

Introduce a symbol $*$ outside the lattice and define the indexed record

$$Z_{n,i} = \begin{cases} X_{n,i}, & X_{n,i} \in A_n, \\ *, & X_{n,i} \notin A_n, \end{cases} \qquad \mathsf{R}_n = (Z_{n,1}, \dots, Z_{n,n}). \tag{2.3}$$

Thus the record reveals both the positions and the exact values of all observations in $A_n$. Set $S_n = \sum_i X_{n,i}$.

**Theorem 2.1** (Rare-record comparison)**.** *Under the assumptions above, for every fixed $C > 0$,*

$$\sup_{m \in \mathcal{C}_n(C)} d_{\mathrm{TV}}\big(\mathcal{L}(\mathsf{R}_n \mid S_n = m), \mathcal{L}(\mathsf{R}_n)\big) \longrightarrow 0, \tag{2.4}$$

*where*

$$\mathcal{C}_n(C) = \{m \in na + d\mathbb{Z} : |m - n\mu_n| \leq C\sqrt{n},\ \mathbb{P}(S_n = m) > 0\}.$$

There is no assumption on the size of $nq_n$. The record may be empty with high probability, contain a tight number of observations, or contain a diverging number.

**Corollary 2.2** (Statistics, marks, and conditioning sets)**.** *The conclusion of Theorem 2.1 holds after applying any measurable map, possibly depending on $n$, to the record. It also holds after adjoining auxiliary marks whose conditional law given the record is unchanged by the sum constraint. Finally, for any sets $B_n \subset \mathcal{C}_n(C)$ with $\mathbb{P}(S_n \in B_n) > 0$,*

$$d_{\mathrm{TV}}\big(\mathcal{L}(\mathsf{R}_n \mid S_n \in B_n), \mathcal{L}(\mathsf{R}_n)\big) \to 0.$$

*Proof.* Total variation contracts under measurable maps and, more generally, under a common Markov kernel. The last assertion follows by writing the conditional law as a mixture over $m \in B_n$ and using the uniform bound in (2.4). □

**Corollary 2.3** (Joint total-variation independence)**.** *Under the hypotheses of Theorem 2.1,*

$$d_{\mathrm{TV}}\big(\mathcal{L}(\mathsf{R}_n, S_n), \mathcal{L}(\mathsf{R}_n) \otimes \mathcal{L}(S_n)\big) \to 0.$$

*Proof.* Disintegrating over the countable variable $S_n$, the distance is the mean of $d_{\mathrm{TV}}(\mathcal{L}(\mathsf{R}_n \mid S_n = m), \mathcal{L}(\mathsf{R}_n))$. The part with $|m - n\mu_n| \leq C\sqrt{n}$ tends to zero uniformly by Theorem 2.1. The remainder is at most $\mathbb{P}(|S_n - n\mu_n| > C\sqrt{n}) \leq \sigma_n^2/C^2$. First let $n \to \infty$, then let $C \to \infty$. □

This integrated conclusion does not replace the pointwise theorem: small average error alone would not control conditioning on a specified lattice value.

**Corollary 2.4** (Uniformity over rare sets)**.** *Fix the row laws of Theorem 2.1. For every sequence $\varepsilon_n \downarrow 0$, the convergence in* (2.4) *is uniform over all deterministic $A \subset a + d\mathbb{Z}$ satisfying $P_n(A) \leq \varepsilon_n$.*

*Proof.* Failure would give a subsequence and a choice of a violating set at each of its indices. Complete these choices by empty sets at the other indices and apply Theorem 2.1. This is a sequential uniformity statement, not a finite-sample rate. □

## 3. A uniform local central limit theorem

The local estimate is classical. We give the uniform version needed for the random number of unrevealed observations. Its moment hypothesis agrees with the uniform-square-integrability extension in Janson [9, Lemma 14.1 and Remark 14.2]; these are numbered 13.1 and 13.2 in the arXiv version.

**Lemma 3.1** (Uniform row and sample-size local limit)**.** *Suppose that $W_n$ satisfies assumptions* (i)–(iii) *of Section 2, with mean $b_n$, variance $s_n^2 \to s^2 > 0$, and limit of maximal span $d$. Let $W_{n,1}, W_{n,2}, \dots$ be independent copies. For each fixed $\eta \in (0,1)$,*

$$\sup_{\substack{\eta n \le N \le n \\ \ell \in Na + d\mathbb{Z}}} \left| \sqrt{N}\mathbb{P}\left( \sum_{i=1}^{N} W_{n,i} = \ell \right) - \frac{d}{s_n\sqrt{2\pi}} \exp\left\{ -\frac{(\ell - Nb_n)^2}{2Ns_n^2} \right\} \right| \to 0. \tag{3.1}$$

*In particular, all such probabilities are at most $K/\sqrt{N}$ for large $n$, with a constant independent of $N$ and $\ell$.*

*Proof.* Replacing $W_n$ by $(W_n - a)/d$ reduces the proof to $a = 0, d = 1$. The moment assumptions and weak convergence imply convergence of the first two moments. Also, weak convergence on the integers implies convergence in total variation: on a finite set the point masses converge, and the remaining masses can be made uniformly small by tightness.

Let $\phi_n(t) = \mathbb{E}e^{itW_n}$ and $\psi_n(t) = e^{-itb_n}\phi_n(t)$. Uniform square integrability, also valid after centering, gives the uniform expansion

$$\psi_n(t) = 1 - \frac{1}{2}s_n^2 t^2 + t^2\varepsilon_n(t), \qquad \lim_{\delta \downarrow 0} \sup_n \sup_{0<|t|\le\delta} |\varepsilon_n(t)| = 0. \tag{3.2}$$

To justify this without a third moment, put $Y_n = W_n - b_n$. For $|Y_n| \le M$, the third-order Taylor remainder divided by $t^2$ is bounded by $|t|MY_n^2/6$. For $|Y_n| > M$, it is bounded by a constant times $Y_n^2$. First choose $M$ using uniform integrability, then let $t \to 0$. The linear term vanishes since $\mathbb{E}Y_n = 0$.

Since $s_n^2$ stays bounded away from zero, (3.2) implies that, for some $\delta, c > 0$ and all sufficiently large $n$,

$$|\phi_n(t)| = |\psi_n(t)| \le e^{-ct^2}, \qquad |t| \le \delta. \tag{3.3}$$

Indeed, squaring the modulus in (3.2) gives $1 - s_n^2 t^2 + o(t^2)$ uniformly.

Let $\phi$ be the characteristic function of the limit law. Maximal span one implies $|\phi(t)| < 1$ for $0 < |t| \le \pi$. Otherwise $e^{itW}$ would be constant on the support and that support would lie in a proper sublattice. Compactness and uniform convergence $\phi_n \to \phi$ therefore give a $\rho < 1$ such that

$$\sup_{\delta \le |t| \le \pi} |\phi_n(t)| \le \rho \tag{3.4}$$

for all large $n$.

For any sequence $N = N(n)$ with $\eta n \le N \le n$, Fourier inversion, with $u = t\sqrt{N}$, gives

$$\sqrt{N}\mathbb{P}\left( \sum_{i=1}^{N} W_{n,i} = \ell \right) = \frac{1}{2\pi} \int_{-\pi\sqrt{N}}^{\pi\sqrt{N}} e^{-iu(\ell - Nb_n)/\sqrt{N}} \psi_n(u/\sqrt{N})^N \, du. \tag{3.5}$$

For each fixed $u$, (3.2) gives

$$\psi_n(u/\sqrt{N})^N - e^{-s_n^2 u^2/2} \longrightarrow 0.$$

On $|u| \le \delta\sqrt{N}$, (3.3) supplies the integrable bound $e^{-cu^2}$. The integral on $\delta\sqrt{N} \le |u| \le \pi\sqrt{N}$ is at most $2\pi\sqrt{N}\rho^N$, by (3.4). Thus the difference between (3.5) and the Gaussian Fourier integral tends to zero uniformly in $\ell$, because the phase factor has modulus one. The Gaussian integral equals the second term in (3.1) with $d = 1$.

If the convergence were not uniform over $N \in [\eta n, n]$, a sequence of violating $N$ could be selected, contradicting the argument just given. Rescaling restores $d$ and proves the lemma. □

## 4. Proof of the rare-record theorem

Write

$$a_n = \mathbb{E}[X_{n,1}\mathbf{1}_{\{X_{n,1}\in A_n\}}], \qquad \beta_n = \mathbb{E}[X_{n,1}^2\mathbf{1}_{\{X_{n,1}\in A_n\}}]. \tag{4.1}$$

For each $M$, $\beta_n \le M^2 q_n + \mathbb{E}[X_{n,1}^2\mathbf{1}_{\{|X_{n,1}|>M\}}]$. Consequently,

$$\beta_n \to 0, \qquad |a_n| \le \sqrt{q_n\beta_n} \to 0. \tag{4.2}$$

For large $n$, define

$$\zeta_n \sim \mathcal{L}(X_{n,1} \mid X_{n,1} \notin A_n), \qquad b_n = \mathbb{E}\zeta_n = \frac{\mu_n - a_n}{1-q_n}, \qquad s_n^2 = \mathrm{Var}(\zeta_n). \tag{4.3}$$

Its law differs from $P_n$ in total variation by $q_n$. Moreover, for nonnegative $g$, $\mathbb{E}g(\zeta_n) \le (1-q_n)^{-1}\mathbb{E}g(X_{n,1})$. Thus $\zeta_n \Rightarrow X$, its squares are uniformly integrable, and $b_n \to \mu$, $s_n^2 \to \sigma^2$. Its support remains in the same lattice. Lemma 3.1 applies both to the original row laws and to these low laws.

**Lemma 4.1** (Exact conditional density)**.** *Let $z$ be a record with positive unconditional probability, with $r$ revealed entries whose sum is $k$. For every $m$ with $\mathbb{P}(S_n = m) > 0$,*

$$D_{n,m}(z) := \frac{\mathbb{P}(\mathsf{R}_n = z \mid S_n = m)}{\mathbb{P}(\mathsf{R}_n = z)} = \frac{\mathbb{P}(\zeta_{n,1} + \cdots + \zeta_{n,n-r} = m-k)}{\mathbb{P}(S_n = m)}. \tag{4.4}$$

*The $\zeta_{n,i}$ in the numerator are independent. An empty sum is zero. In particular, $D_{n,m} \ge 0$ and $\mathbb{E}D_{n,m}(\mathsf{R}_n) = 1$.*

*Proof.* Given the indexed record, the unrevealed coordinates are independent with law (4.3); independence follows by factoring the original product law coordinate by coordinate. Their sum must equal $m-k$. Bayes' formula proves (4.4). There is no combinatorial factor, because the revealed positions are part of the record. Summing the conditional probabilities gives the normalization. □

*Remark* 4.2 (Sufficiency of the revealed count and sum). Write $R_n$ for the number of revealed observations and $K_n$ for their sum. Since (4.4) depends on the record only through $(R_n, K_n)$,

$$d_{\mathrm{TV}}(\mathcal{L}(\mathsf{R}_n \mid S_n = m), \mathcal{L}(\mathsf{R}_n)) = d_{\mathrm{TV}}(\mathcal{L}((R_n,K_n) \mid S_n = m), \mathcal{L}(R_n,K_n)).$$

Indeed, the sign of $D_{n,m} - 1$ is constant on each fiber of $(R_n, K_n)$, so summing the absolute differences over each fiber gives equality. This is the standard sufficiency reduction for total variation discussed in [3, Theorem 2]. In particular, retaining the original labels does not incur an additional approximation error.

The following elementary form of Scheffé's argument [12] supplies total variation without a separate uniform-integrability estimate for the likelihood ratios.

**Lemma 4.3** (One-sided density identity)**.** *If $D \ge 0$ and $\mathbb{E}D = 1$, then*

$$\frac{1}{2}\mathbb{E}|D-1| = \mathbb{E}(1-D)_+. \tag{4.5}$$

*Consequently, for normalized nonnegative densities $D_n$ on possibly different probability spaces, $D_n \to 1$ in probability implies $\mathbb{E}|D_n - 1| \to 0$.*

*Proof.* Since $\mathbb{E}(D-1) = 0$, its positive and negative parts have equal expectations. Also $0 \le (1-D_n)_+ \le 1$. A bounded sequence converging to zero in probability converges to zero in expectation. □

*Proof of Theorem 2.1.* Let

$$R_n = \sum_{i=1}^n \mathbf{1}_{\{X_{n,i}\in A_n\}}, \qquad K_n = \sum_{i=1}^n X_{n,i}\mathbf{1}_{\{X_{n,i}\in A_n\}}. \tag{4.6}$$

Independence and (4.2) give

$$\mathrm{Var}\, R_n \le nq_n = o(n), \qquad \mathrm{Var}\, K_n \le n\beta_n = o(n). \tag{4.7}$$

In particular, $R_n/n \xrightarrow{\mathbb{P}} 0$, and

$$U_n := (K_n - na_n) - (R_n - nq_n)b_n = o_{\mathbb{P}}(\sqrt{n}).$$

The exact centering identity is

$$(m - K_n) - (n - R_n)b_n = (m - n\mu_n) - U_n. \tag{4.8}$$

Here the first term on the right is allowed to be $O(\sqrt{n})$, not necessarily $o(\sqrt{n})$.

On an event of probability tending to one, $N_n = n - R_n \in [n/2, n]$. Set

$$t_{n,m} = \frac{m - n\mu_n}{\sigma_n\sqrt{n}}, \qquad v_{n,m} = \frac{(m - K_n) - N_n b_n}{s_n\sqrt{N_n}}.$$

Equation (4.8), $N_n/n \xrightarrow{\mathbb{P}} 1$ and $s_n/\sigma_n \to 1$ imply

$$\sup_{m \in \mathcal{C}_n(C)} |v_{n,m} - t_{n,m}| \xrightarrow{\mathbb{P}} 0. \tag{4.9}$$

Indeed, the part containing $m - n\mu_n$ is bounded uniformly by a constant times $|\sigma_n\sqrt{n}/(s_n\sqrt{N_n}) - 1|$, and the remaining part is bounded by $|U_n|/(s_n\sqrt{N_n})$.

Let $\varphi(t) = (2\pi)^{-1/2}e^{-t^2/2}$. Lemma 3.1 yields, uniformly over the central targets,

$$\mathbb{P}(S_n = m) = \frac{d}{\sigma_n\sqrt{n}}\{\varphi(t_{n,m}) + o(1)\}, \tag{4.10}$$

$$\mathbb{P}\left(\sum_{i=1}^{N_n} \zeta_{n,i} = m - K_n\right) = \frac{d}{s_n\sqrt{N_n}}\{\varphi(v_{n,m}) + o(1)\} \tag{4.11}$$

on $N_n \geq n/2$. The errors are deterministic and uniform in the possible values of $N_n, K_n, m$. Lattice compatibility holds: $m \in na + d\mathbb{Z}$ and $K_n \in R_n a + d\mathbb{Z}$, so $m - K_n \in N_n a + d\mathbb{Z}$.

The values $t_{n,m}$ range over a bounded interval. Thus $\varphi(t_{n,m})$ is uniformly bounded away from zero, and (4.10) is at least $c_C/\sqrt{n}$ for large $n$. Combining (4.4) and (4.9)–(4.11) gives

$$\sup_{m \in \mathcal{C}_n(C)} |D_{n,m}(\mathsf{R}_n) - 1| \xrightarrow{\mathbb{P}} 0. \tag{4.12}$$

Values on the event $N_n < n/2$ do not affect convergence in probability.

By Lemmas 4.1 and 4.3,

$$\begin{aligned}\sup_{m \in \mathcal{C}_n(C)} d_{\mathrm{TV}}(\mathcal{L}(\mathsf{R}_n \mid S_n = m), \mathcal{L}(\mathsf{R}_n)) &= \sup_m \mathbb{E}(1 - D_{n,m}(\mathsf{R}_n))_+ \\ &\leq \mathbb{E}\sup_m (1 - D_{n,m}(\mathsf{R}_n))_+ \longrightarrow 0.\end{aligned}$$

The integrand is bounded by one and converges to zero in probability by (4.12). This proves the theorem. □

## 5. Upper order statistics

**Theorem 5.1** (Growing ranks for triangular arrays)**.** *Assume the hypotheses of Theorem 2.1, and suppose that the row laws are nonnegative and that the limit law has unbounded support. For any $1 \leq k_n = o(n)$, the conclusion* (1.2) *holds with $X_{(j)}$ replaced by the decreasing order statistics of $X_{n,1}, \ldots, X_{n,n}$ and $\mu$ replaced by $\mu_n$.*

*Proof.* Put $u_n = \sqrt{k_n/n} \to 0$. For all large $n$, choose the largest integer $h_n \geq 0$ for which $\mathbb{P}(X_{n,1} \geq h_n) \geq u_n$. Such an integer exists and is finite. For each fixed $h$, weak convergence gives $\mathbb{P}(X_{n,1} \geq h) \to \mathbb{P}(X \geq h) > 0$, so $h_n \to \infty$. Uniform tightness then implies $q_n := \mathbb{P}(X_{n,1} \geq h_n) \to 0$. Moreover,

$$\frac{nq_n}{k_n} \geq \sqrt{\frac{n}{k_n}} \to \infty. \tag{5.1}$$

The number $R_n$ of observations at or above the threshold is binomial. For large $n$, $k_n \le nq_n/2$, and Chebyshev's inequality gives

$$\mathbb{P}(R_n < k_n) \le \frac{nq_n}{(nq_n - k_n)^2} \le \frac{4}{nq_n} \to 0.$$

Theorem 2.1 transfers this conclusion uniformly to the conditioned laws. Sort the revealed values, retain the first $k_n$, and fill with zeros if fewer than $k_n$ are available. This is a measurable function of the record and agrees with the actual rank vector on $R_n \ge k_n$. Total-variation contraction and the two failure probabilities prove the assertion. □

*Proof of Theorem 1.1.* For an unbounded law, take the row law constant in Theorem 5.1. If a fixed law has finite maximum support value $b$, then $\mathbb{P}(X = b) > 0$, so fewer than $k_n = o(n)$ occurrences of $b$ has probability at most $e^{-cn}$ for large $n$. The conditional probability of this failure is at most $e^{-cn}/\mathbb{P}(S_n = m) = O(\sqrt{n}e^{-cn})$, uniformly in the central window by Lemma 3.1. Both rank vectors therefore equal $(b, \dots, b)$ with probability tending to one. □

## 6. Weighted allocations and conditioned trees

Let $w = (w_j)_{j\ge 0}$ be nonnegative weights with $w_0 > 0$ and unbounded support. Define

$$\Phi(t) = \sum_{j\ge 0} w_j t^j, \qquad \Psi(t) = \frac{t\Phi'(t)}{\Phi(t)}, \qquad \rho = \text{radius of convergence of } \Phi.$$

For $0 < t \le \rho$ with $\Phi(t) < \infty$, write

$$p_t(j) = \frac{w_j t^j}{\Phi(t)}. \tag{6.1}$$

At $t = \rho$, moment assumptions are understood as convergent sums. Let $\nu = \lim_{t\uparrow\rho} \Psi(t)$. An allocation $Y = (Y_1, \dots, Y_n)$ of $m$ balls is distributed by

$$\mathbb{P}(Y = y) = \frac{\prod_{i=1}^n w_{y_i}}{Z_{n,m}}, \qquad Z_{n,m} = \sum_{\substack{y_i \ge 0 \\ \sum_i y_i = m}} \prod_{i=1}^n w_{y_i} > 0. \tag{6.2}$$

The variables with law $p_t$ will be denoted $\xi_t$.

**Lemma 6.1** (Conditional representation)**.** *For every allowable $t$, the law in* (6.2) *equals the law of $n$ independent copies of $\xi_t$ conditioned to sum to $m$.*

*Proof.* On $\sum_i y_i = m$, the product of the probabilities in (6.1) is $t^m \Phi(t)^{-n} \prod_i w_{y_i}$. The first factor is constant over the conditioning event and cancels on normalization. □

**Theorem 6.2** (Allocation extremes)**.** *Suppose $\Psi(\tau) = \lambda > 0$ and $\mathrm{Var}(\xi_\tau) \in (0, \infty)$.*

(i) *If $m_n = \lambda n + O(\sqrt{n})$, then for every $k_n = o(n)$,*

$$d_{\mathrm{TV}}\big(\mathcal{L}((Y_{(j)})_{j\le k_n}), \mathcal{L}((\xi_{\tau,(j)})_{j\le k_n})\big) \to 0, \tag{6.3}$$

*along admissible allocations.*

(ii) *If $0 < \lambda < \nu$ and $m_n/n \to \lambda$, choose $\tau_n$ by $\Psi(\tau_n) = m_n/n$. Then* (6.3) *holds for every $k_n = o(n)$ with $\xi_\tau$ replaced by $\xi_{\tau_n}$.*

(iii) *The conclusion in* (ii) *also holds when $\lambda = \nu < \infty$, $m_n/n \le \nu$, and the boundary law $p_\rho$ has finite positive variance.*

*Proof.* Part (i) follows from Lemma 6.1 and Theorem 1.1. The common lattice is $d\mathbb{Z}$, where $d = \gcd\{j : w_j > 0\}$; since $w_0 > 0$, this is the maximal lattice span.

For (ii), $\Psi$ is strictly increasing on $(0, \rho)$, because differentiation with respect to $\log t$ gives $\mathrm{Var}(\xi_t) > 0$. Thus $\tau_n \to \tau < \rho$. Choose $t_* \in (\tau, \rho)$ with $\tau_n \le t_*$ for all large $n$. Since $\Phi(\tau_n) \ge w_0$,

$$p_{\tau_n}(j) \le w_0^{-1} w_j t_*^j.$$

The dominating sequence has finite second moment, indeed an exponential moment. The row laws converge pointwise to $p_\tau$, are uniformly square-integrable and share the same support. Their means are exactly $m_n/n$. Apply Theorem 5.1 and Lemma 6.1.

For (iii), $\tau_n \le \rho$ and $\tau_n \to \rho$. Monotone convergence gives $\Phi(\tau_n) \to \Phi(\rho)$, and, for large $n$,

$$p_{\tau_n}(j) \le 2p_\rho(j).$$

Finite second moment of $p_\rho$ supplies uniform integrability. The same argument applies, including indices at which $\tau_n = \rho$. □

**Corollary 6.3** (The fixed-rank question)**.** *The fixed-law and moving-tilt conclusions proposed in Janson's Problem 19.10 hold jointly for every fixed number of upper ranks. Under the central fixed-law centering, and under the mean-matched moving tilts in Theorem 6.2, the number of ranks may be any $o(n)$ sequence.*

*Proof.* Take $k_n = J$ in Theorem 6.2. The weaker subcritical fixed-law centering included in the source theorem is established in Theorem 7.2 below, so it is not being inferred from the central-window theorem alone. □

**Corollary 6.4** (Ranked outdegrees)**.** *Let $T_n$ be a Galton–Watson plane tree with a critical nondegenerate offspring law $X$ of finite variance, conditioned to have $n$ vertices. For every $k_n = o(n)$, the decreasing vector of its first $k_n$ outdegrees and the first $k_n$ ranks of $n$ independent copies of $X$ have total-variation distance tending to zero, along admissible $n$.*

*Proof.* For a nonnegative degree sequence summing to $n-1$, put $s_0 = 0$ and $s_j = \sum_{i=1}^{j}(x_i - 1)$, so $s_n = -1$. Let $j$ be the first location of the minimum of $s_0, \dots, s_{n-1}$. Starting the sequence at $j+1$ gives nonnegative partial sums until the last step: before the wrap they are $s_k - s_j \ge 0$, and after the wrap they are $-1 - s_j + s_k \ge 0$, since $k < j$ implies $s_k \ge s_j + 1$. When $j > 0$, the intermediate sum at the wrap is $-1 - s_j \ge 0$ because $s_j \le -1$. Conversely, these inequalities force any valid starting position to follow that first minimum. Thus exactly one of the $n$ cyclic starting positions gives a depth-first outdegree sequence of a plane tree. The total $-1$ rules out a nontrivial period of the increment sequence. This is the elementary cyclic argument underlying the total-progeny identity [6]; see also [9, Section 15].

The product probability $\prod_i \mathbb{P}(X = x_i)$ is invariant under cyclic shifts. Consequently the multiset of outdegrees of $T_n$ has the same distribution as the multiset of $n$ independent offspring variables conditioned to sum to $n-1$; this is the usual allocation representation, also discussed in [9, Section 15]. Since $\mathbb{E}X = 1$, the target differs from the mean by $-1$. Apply Theorem 1.1, or its bounded-support observation. □

*Remark* 6.5. The tree corollary concerns ranked outdegrees and other cyclically invariant statistics. It does not identify depth-first vertex positions with the independent labels of the rare record. No claim about those positions is made.

## 7. Weaker centering in the subcritical case

The fixed-law comparison at $m_n/n = \lambda + o(1/\log n)$ cannot be obtained by replacing the central Gaussian approximation with its value at zero. We instead compare the two independent rank vectors after a small exponential tilt. The following lemma handles large atoms at the threshold.

**Lemma 7.1** (Censored comparison of upper ranks)**.** *Let $p$ be a fixed law of unbounded support on $\mathbb{Z}_{\ge 0}$, and let $\widetilde{p}_n$ have the same support. Suppose there are integers $B_n$ and numbers $\epsilon_n \to 0$ such that*

$$n\{p((B_n,\infty)) + \widetilde{p}_n((B_n,\infty))\} \to 0, \tag{7.1}$$

$$\sup_{\substack{0\le j\le B_n \\ p(j)>0}} \left|\frac{\widetilde{p}_n(j)}{p(j)} - 1\right| \le \epsilon_n. \tag{7.2}$$

*If $k_n = o(n)$ and $k_n\epsilon_n^2 \to 0$, then the first $k_n$ upper ranks of $n$ independent $p$ variables and $n$ independent $\widetilde{p}_n$ variables have total-variation distance tending to zero.*

*Proof.* Choose numbers $L_n$ such that

$$L_n \to \infty, \qquad k_n = o(L_n), \qquad L_n = o(n), \qquad L_n \epsilon_n^2 \to 0. \tag{7.3}$$

Such a choice exists because $k_n/n \to 0$ and $k_n \epsilon_n^2 \to 0$: multiply $\max(k_n, 1)$ by a sequence tending to infinity sufficiently slowly.

Let $a_n$ be the largest integer in $\operatorname{supp} p$ for which $np([a_n, \infty)) \geq L_n$. Then

$$np((a_n, \infty)) < L_n \leq np([a_n, \infty)). \tag{7.4}$$

To verify the strict inequality, let $a_n'$ be the next support point. Its tail is $p([a_n', \infty)) = p((a_n, \infty))$, and maximality gives the claim. Also $a_n \to \infty$, and (7.1) ensures $a_n \leq B_n$ for large $n$.

Censor each independent observation by keeping its value only in $(a_n, B_n]$ and replacing everything else by $*$. Write $Q_n$ and $\widetilde{Q}_n$ for the one-coordinate censored laws and $r_n = p((a_n, B_n]) < L_n/n$. With

$$H^2(Q, \widetilde{Q}) := \sum_z (\sqrt{Q\{z\}} - \sqrt{\widetilde{Q}\{z\}})^2,$$

(7.2) gives a contribution at most $\epsilon_n^2 r_n$ from the retained values. The difference of the two cemetery masses has modulus at most $\epsilon_n r_n$. Since both cemetery masses tend to one, their contribution is at most $2\epsilon_n^2 r_n^2$ for large $n$. Therefore

$$H^2(Q_n, \widetilde{Q}_n) \leq 2\epsilon_n^2 r_n, \qquad d_{\mathrm{TV}}(Q_n^{\otimes n}, \widetilde{Q}_n^{\otimes n}) \leq \sqrt{nH^2(Q_n, \widetilde{Q}_n)} \leq \sqrt{2L_n}\epsilon_n \to 0. \tag{7.5}$$

Here tensorization follows by multiplying Hellinger affinities, and $d_{\mathrm{TV}} \leq H$ follows from Cauchy–Schwarz.

Under $p$, the expected number of observations at least $a_n$ is at least $L_n$. Under $\widetilde{p}_n$, the expected number in $[a_n, B_n]$ is at least

$$(1 - \epsilon_n)np([a_n, B_n]) \geq (1 - \epsilon_n)(L_n - o(1)).$$

Binomial concentration and (7.3) show that both samples have at least $k_n$ observations at least $a_n$ with probability tending to one. By (7.1), neither sample exceeds $B_n$ with probability tending to one.

On these events, its first $k_n$ ranks are obtained by sorting the retained values and filling any remaining positions with $a_n$. The same map applies to both censored records. Combining (7.5) with the failure probabilities proves the result. The atom at $a_n$ is never compared coordinatewise; it is supplied by the fill operation, which is why no bound on that atom is needed. □

**Theorem 7.2** (Weak subcritical centering)**.** *In the setting of Theorem 6.2, suppose* $0 < \lambda < \nu$ *and put* $\delta_n = m_n/n - \lambda$*. If* $\delta_n \log n \to 0$*, then* (6.3) *holds for every fixed number of ranks. More generally, it holds for any* $k_n$ *satisfying*

$$k_n = o(n), \qquad k_n(\delta_n \log n)^2 \longrightarrow 0. \tag{7.6}$$

*Proof.* Take the mean-matched tilt $\tau_n$ in Theorem 6.2(ii). Let $p = p_\tau$, $\theta_n = \log(\tau_n/\tau)$ and $M(\theta) = \mathbb{E}_p e^{\theta X}$. Then

$$p_{\tau_n}(j) = p(j)\frac{e^{\theta_n j}}{M(\theta_n)}. \tag{7.7}$$

Since the derivative of the tilted mean at zero is $\operatorname{Var}_p X > 0$, the inverse function theorem gives $\theta_n = O(\delta_n)$. The fixed law has a positive exponential moment because $\tau < \rho$. For some $t > 0$,

$$\sup_n \mathbb{E}_{p_{\tau_n}} e^{tX} < \infty$$

after discarding finitely many indices. Choose $B_n = \lceil C_0 \log n \rceil$ with $C_0 t > 2$. Markov's inequality yields (7.1).

For $j \leq B_n$, (7.7) and smoothness of $\log M$ give

$$\left|\log \frac{p_{\tau_n}(j)}{p(j)}\right| \leq |\theta_n| B_n + |\log M(\theta_n)| = O(|\delta_n| \log n) = o(1).$$

Hence (7.2) holds with $\epsilon_n \le C_1|\delta_n|\log n$ for large $n$. Lemma 7.1 compares the independent rank vectors under $p_{\tau_n}$ and $p_\tau$. Theorem 6.2(ii) compares the allocation ranks to the former. The triangle inequality proves the theorem. □

*Remark* 7.3. The restriction in (7.6) is a sufficient condition in the weak-centering regime, not a claimed optimal growing-rank range there. In the central fixed-law regime, Theorem 1.1 gives every $k_n = o(n)$ without that extra restriction.

## 8. Poissonized rare counts and upper ranks

The indexed record contains more information than is needed to describe its multiset of values. After the original sample labels are forgotten, a further independent-Poisson approximation is available without requiring a bounded expected number of rare observations.

For $x \in A_n$, put

$$C_{n,x} = \sum_{i=1}^{n} \mathbf{1}_{\{X_{n,i}=x\}}, \qquad \mathsf{C}_n = (C_{n,x})_{x\in A_n}.$$

Let $\mathsf{Z}_n = (Z_{n,x})_{x\in A_n}$ have independent coordinates

$$Z_{n,x} \sim \mathrm{Poisson}(nP_n\{x\}).$$

Both count vectors have finite total mass almost surely, since $nq_n < \infty$ for each $n$.

**Theorem 8.1** (Poisson approximation of all rare-value counts)**.** *Under the hypotheses of Theorem 2.1, define*

$$e_n(C) = \sup_{m\in\mathcal{C}_n(C)} d_{\mathrm{TV}}(\mathcal{L}(\mathsf{R}_n \mid S_n = m), \mathcal{L}(\mathsf{R}_n)).$$

*Then*

$$\sup_{m\in\mathcal{C}_n(C)} d_{\mathrm{TV}}(\mathcal{L}(\mathsf{C}_n \mid S_n = m), \mathcal{L}(\mathsf{Z}_n)) \le e_n(C) + (1 - e^{-nq_n})q_n \longrightarrow 0. \tag{8.1}$$

*In particular, the assertion holds when $nq_n \to \infty$.*

*Proof.* The first comparison, between conditioned and unconditioned $\mathsf{C}_n$, follows from Theorem 2.1 by contraction. Under the independent sample law, the total number of rare observations has distribution $\mathrm{Bin}(n, q_n)$. Given that total, their values are independent marks with law $P_n(\,\cdot \mid A_n)$. The total count of $\mathsf{Z}_n$ has distribution $\mathrm{Poisson}(nq_n)$, and, conditional on the total, its marks have exactly the same law. These are the usual marking and splitting properties of independent Poisson counts.

Applying the common marking kernel gives

$$d_{\mathrm{TV}}(\mathcal{L}(\mathsf{C}_n), \mathcal{L}(\mathsf{Z}_n)) \le d_{\mathrm{TV}}(\mathrm{Bin}(n, q_n), \mathrm{Poisson}(nq_n)).$$

Projection onto the total count gives the reverse inequality. Thus the two distances are equal. The standard Poisson approximation bound [4, 2] is

$$d_{\mathrm{TV}}(\mathrm{Bin}(n, q), \mathrm{Poisson}(nq)) \le (1 - e^{-nq})q, \qquad 0 \le q \le 1. \tag{8.2}$$

For example, Theorem 1 of [2], whose total-variation convention is twice ours, gives (8.2) by taking independent Bernoulli indicators: $b_1 = nq^2$ and $b_2 = b_3 = 0$. The case $q = 0$ is immediate. The triangle inequality proves (8.1). □

The distinction between values and labels is important. Theorem 8.1 concerns counts indexed by the possible observation values. It does not replace each original sample position by an independent Poisson count. The latter process can put several observations at one position and is not the approximation used here.

**Corollary 8.2** (Poissonized upper ranks)**.** *In Theorem 5.1, let $M_n \sim \mathrm{Poisson}(n)$ be independent of a sequence of independent $P_n$ observations. Write $\widehat{X}_{n,(1)} \ge \widehat{X}_{n,(2)} \ge \cdots$ for their decreasing order statistics up to $M_n$, padding with zeros when necessary. For every $k_n = o(n)$,*

$$\sup_{m\in\mathcal{C}_n(C)} d_{\mathrm{TV}}\left(\mathcal{L}((X_{n,(j)})_{j\le k_n} \mid S_n = m), \mathcal{L}((\widehat{X}_{n,(j)})_{j\le k_n})\right) \to 0.$$

*Proof.* Choose the upper threshold $h_n$ from the proof of Theorem 5.1. Its tail probability $q_n$ tends to zero and $nq_n/k_n \to \infty$. The corresponding rare counts of the Poissonized sample are the independent counts in Theorem 8.1. Their total is Poisson with mean $nq_n$, so Chebyshev's inequality gives probability tending to one of at least $k_n$ retained observations. The independent fixed-size sample has the same property, and Theorem 2.1 transfers it to every conditioned sample in the central window. Sort each rare multiset and take its first $k_n$ entries, padding by zeros. The same map recovers both rank vectors outside events of probability tending uniformly to zero. Apply Theorem 8.1 and contraction. □

## 9. Several constraints and a growing coordinate block

The same argument allows finitely many sum constraints and the simultaneous observation of a prescribed sublinear set of coordinates. The conclusion is joint: separate approximations for a coordinate block and a rare record would not imply it. Conditional representations with vector weights are already part of the framework of [3]; the theorem below supplies the finite-variance asymptotic for this particular observation scheme.

Fix $r \geq 1$. For each $n$, let $V_{n,1}, \dots, V_{n,n}$ be independent with common law $Q_n$ on $\mathbb{Z}^r$, mean $\mu_n$ and covariance matrix $\Sigma_n$. Assume

(i) $V_{n,1} \Rightarrow V$, where $\Sigma = \mathrm{Cov}(V)$ is positive definite;
(ii) $\{\|V_{n,1}\|^2 : n \geq 1\}$ is uniformly integrable;
(iii) the additive subgroup generated by

$$\{x - y : x, y \in \operatorname{supp} \mathcal{L}(V)\} \tag{9.1}$$

is $\mathbb{Z}^r$.

Here and below $\|\cdot\|$ is Euclidean norm. The mean and covariance converge to those of $V$. Assumption (iii) is the full-lattice condition; it is stronger than saying only that the support is not contained in an affine hyperplane. A fixed affine lattice of full rank can be reduced to this setting by a change of coordinates.

Let $A_n \subset \mathbb{Z}^r$ satisfy $q_n = Q_n(A_n) \to 0$, and let $I_n \subset \{1, \dots, n\}$ be deterministic with $\ell_n = |I_n| = o(n)$. Observe

$$\mathsf{T}_n = \left((V_{n,i})_{i \in I_n}, (Z_{n,i})_{i \notin I_n}\right), \qquad Z_{n,i} = \begin{cases} V_{n,i}, & V_{n,i} \in A_n, \\ *, & V_{n,i} \notin A_n. \end{cases} \tag{9.2}$$

The order of the coordinates in each component is their original order. Write $S_n = \sum_i V_{n,i}$.

**Theorem 9.1** (Joint block and rare-record comparison)**.** *Under the assumptions above, for every $C > 0$,*

$$\sup_{m \in \mathcal{D}_n(C)} d_{\mathrm{TV}}(\mathcal{L}(\mathsf{T}_n \mid S_n = m), \mathcal{L}(\mathsf{T}_n)) \to 0, \tag{9.3}$$

*where*

$$\mathcal{D}_n(C) = \{m \in \mathbb{Z}^r : \|m - n\mu_n\| \leq C\sqrt{n},\ \mathbb{P}(S_n = m) > 0\}.$$

*The unconditional law in* (9.3) *is the product of the law of the prescribed coordinate block and that of the rare record outside the block.*

We first record the multivariate version of the local estimate. For a positive-definite matrix $\Gamma$, let

$$g_\Gamma(x) = \frac{\exp(-x^{\mathsf{T}}\Gamma^{-1}x/2)}{(2\pi)^{r/2}\sqrt{\det \Gamma}}.$$

**Lemma 9.2** (Multivariate local estimate)**.** *Suppose the row laws of $W_n \in \mathbb{Z}^r$ satisfy* (i)–(iii) *above, with means $b_n$ and covariance matrices $\Gamma_n \to \Gamma > 0$. For every fixed $\eta \in (0, 1)$,*

$$\sup_{\substack{\eta n \leq N \leq n \\ z \in \mathbb{Z}^r}} \left| N^{r/2}\mathbb{P}\left(\sum_{i=1}^N W_{n,i} = z\right) - g_{\Gamma_n}\left(\frac{z - Nb_n}{\sqrt{N}}\right)\right| \to 0. \tag{9.4}$$

*Proof.* This is the Fourier proof of Lemma 3.1 in $r$ dimensions; we give the details needed for the lattice condition. Uniform integrability of the centered squared norms gives, for $\psi_n(t) = \mathbb{E}e^{it^{\mathsf{T}}(W_n - b_n)}$,

$$\psi_n(t) = 1 - \tfrac{1}{2}t^{\mathsf{T}}\Gamma_n t + o(\|t\|^2) \quad (t \to 0),$$

with a remainder uniform in $n$. Indeed, split according to $\|W_n - b_n\| \le M$ and its complement, as in (3.2). The eigenvalues of $\Gamma_n$ stay bounded away from zero. It follows that $|\psi_n(t)| \le e^{-c\|t\|^2}$ on a fixed neighborhood of zero.

Weak convergence on $\mathbb{Z}^r$ implies total-variation convergence of the one-coordinate laws, hence uniform convergence of their characteristic functions. If the limiting characteristic function has modulus one at $t$, then $t^{\mathsf{T}}(x-y) \in 2\pi\mathbb{Z}$ for all support differences. By (9.1), this forces $t \in 2\pi\mathbb{Z}^r$. On the torus $[-\pi,\pi]^r$, compactness therefore gives a uniform bound $\rho < 1$ outside any fixed neighborhood of zero.

Fourier inversion followed by $u = \sqrt{N}\,t$ now gives

$$N^{r/2}\mathbb{P}\left(\sum_{i=1}^{N} W_{n,i} = z\right) = \frac{1}{(2\pi)^r}\int_{[-\pi\sqrt{N},\pi\sqrt{N}]^r} e^{-iu^{\mathsf{T}}(z - Nb_n)/\sqrt{N}}\psi_n(u/\sqrt{N})^N\,du.$$

On bounded $u$ sets the second factor converges to $\exp(-u^{\mathsf{T}}\Gamma_n u/2)$ with error tending to zero. The small-frequency bound supplies an integrable Gaussian majorant; the remaining integral is at most a constant times $N^{r/2}\rho^N$. Comparison with the Gaussian Fourier integral proves (9.4), uniformly in $z$ because the phase has modulus one. Uniformity over $\eta n \le N \le n$ follows by the same subsequence argument as in Lemma 3.1. □

*Proof of Theorem 9.1.* For large $n$, put

$$a_n = \mathbb{E}[V_{n,1}\mathbf{1}_{\{V_{n,1}\in A_n\}}], \qquad \zeta_n \sim \mathcal{L}(V_{n,1} \mid V_{n,1} \notin A_n), \qquad b_n = \frac{\mu_n - a_n}{1 - q_n}.$$

Uniform square integrability and $q_n \to 0$ imply

$$\beta_n := \mathbb{E}[\|V_{n,1}\|^2\mathbf{1}_{\{V_{n,1}\in A_n\}}] \to 0. \tag{9.5}$$

The complement laws converge to $\mathcal{L}(V)$ in total variation, their squared norms are uniformly integrable, and their covariance matrices $\Gamma_n$ converge to $\Sigma$. Thus Lemma 9.2 applies to both sets of row laws.

Let $R_n$ count rare observations outside $I_n$, let $K_n$ be the sum of all the revealed vectors, and set $N_n = n - \ell_n - R_n$. Given the observation (9.2), the $N_n$ unrevealed variables are independent with law $\zeta_n$. The exact conditional density is therefore

$$D_{n,m}(\mathsf{T}_n) = \frac{\mathbb{P}(\zeta_{n,1} + \cdots + \zeta_{n,N_n} = m - K_n)}{\mathbb{P}(S_n = m)}. \tag{9.6}$$

The empty sum has value zero; as before this density is nonnegative and has expectation one.

Define the centered revealed contribution

$$\begin{aligned} U_n = &\sum_{i\in I_n}(V_{n,i} - \mu_n) \\ &+ \sum_{i\notin I_n}\Big((V_{n,i} - b_n)\mathbf{1}_{\{V_{n,i}\in A_n\}} - (\mu_n - b_n)\Big). \end{aligned} \tag{9.7}$$

Each summand is centered because $a_n - q_n b_n = \mu_n - b_n$. Independence and boundedness of $b_n$ give

$$\begin{aligned} \mathbb{E}\|U_n\|^2 &\le \ell_n \operatorname{tr}\Sigma_n + (n-\ell_n)\mathbb{E}[\|V_{n,1} - b_n\|^2\mathbf{1}_{A_n}(V_{n,1})] \\ &\le \ell_n \operatorname{tr}\Sigma_n + 2n(\beta_n + \|b_n\|^2 q_n) = o(n). \end{aligned} \tag{9.8}$$

Also $N_n/n \xrightarrow{\mathbb{P}} 1$. Direct expansion gives the identity

$$(m - K_n) - N_n b_n = (m - n\mu_n) - U_n. \tag{9.9}$$

Uniformly for $m \in \mathcal{D}_n(C)$, the vectors $(m - n\mu_n)/\sqrt{n}$ and $((m-K_n) - N_n b_n)/\sqrt{N_n}$ differ by $o_{\mathbb{P}}(1)$.

On $N_n \geq n/2$, apply Lemma 9.2 to (9.6). The denominator is bounded below by $c_C n^{-r/2}$, because $\Sigma_n \to \Sigma > 0$ and the central arguments lie in a fixed ball. The numerator has the same Gaussian value up to $o_{\mathbb{P}}(1)$ after multiplication by $n^{r/2}$: use (9.9), $\Gamma_n \to \Sigma$, and $N_n/n \xrightarrow{\mathbb{P}} 1$. The local-limit errors are deterministic and uniform in all possible $N_n, K_n, m$. Consequently

$$\sup_{m \in \mathcal{D}_n(C)} |D_{n,m}(\mathsf{T}_n) - 1| \xrightarrow{\mathbb{P}} 0.$$

The event $N_n < n/2$ has probability tending to zero and does not affect this assertion. The one-sided identity of Lemma 4.3 proves (9.3). Finally, the block and the record outside it are independent before conditioning because they depend on disjoint sets of independent coordinates. □

**Corollary 9.3** (A coordinate block together with the upper ranks)**.** *Let $X$ be a nonnegative integer-valued law of unbounded support and finite positive variance. For deterministic $I_n \subset \{1, \dots, n\}$ with $|I_n| = o(n)$ and $k_n = o(n)$, the joint law*

$$\mathcal{L}\left((X_i)_{i \in I_n}, (X_{(j)})_{j \leq k_n} \,\middle|\, S_n = m\right)$$

*is at total-variation distance tending uniformly to zero from its unconditional law for attainable $|m - n\mathbb{E}X| \leq C\sqrt{n}$.*

*Proof.* Reduce the maximal lattice of $X$ to span one and apply Theorem 9.1 with $r = 1$. Choose a rare upper tail of probability $q_n \to 0$ with $nq_n/k_n \to \infty$, as in Theorem 5.1. Among the $n - |I_n|$ coordinates outside the block there are at least $k_n$ tail observations with probability tending to one. The same holds after conditioning by Theorem 9.1. The revealed block and the rare observations outside it therefore determine the first $k_n$ ranks of the whole sample with high probability. Apply the same sorting map to both laws. No assertion of independence between the block and the ranks is needed or made. □

9.1. **A prescribed number of zeros and of leaves.** Here is a concrete application in which the extra constraint is not redundant. For a fixed nonnegative integer-valued $X$, write $p_j = \mathbb{P}(X = j)$ and

$$S_n = \sum_{i=1}^{n} X_i, \qquad E_n = \sum_{i=1}^{n} \mathbf{1}_{\{X_i = 0\}}.$$

**Corollary 9.4** (Upper ranks under two constraints)**.** *Suppose $p_0 p_1 p_2 > 0$ and $\mathbb{E}X^2 < \infty$. For every $k_n = o(n)$ and $C > 0$,*

$$\sup_{(m,e) \in \mathcal{H}_n(C)} d_{\mathrm{TV}}\left(\mathcal{L}((X_{(j)})_{j \leq k_n} \mid S_n = m, E_n = e), \mathcal{L}((X_{(j)})_{j \leq k_n})\right) \to 0,$$

*where $\mathcal{H}_n(C)$ consists of attainable integer pairs satisfying $|m - n\mathbb{E}X| + |e - np_0| \leq C\sqrt{n}$.*

*Proof.* Take $V = (X, \mathbf{1}_{\{X=0\}})$. Its support contains $(0,1), (1,0), (2,0)$, whose differences generate $\mathbb{Z}^2$. These three points are not collinear, so the covariance matrix is positive definite. The second moment is finite. For unbounded $X$, apply Theorem 9.1 with an empty block and the upper-tail set $A_n = \{(x, b) : x \geq h_n\}$ used in Theorem 5.1. The same binomial-tail and sorting argument proves the result. If $X$ has a finite maximum support value, the probability of fewer than $k_n$ copies of that value is exponentially small. Lemma 9.2 bounds the conditioning probability below by $c_C/n$, uniformly on $\mathcal{H}_n(C)$, so the conditional failure probability still tends to zero. □

**Corollary 9.5** (Ranked outdegrees with a prescribed leaf count)**.** *Let $X$ be a critical offspring law with $p_0 p_1 p_2 > 0$ and finite variance. Condition a Galton–Watson plane tree to have $n$ vertices and $e_n$ leaves, where $|e_n - np_0| = O(\sqrt{n})$ and the conditioning event has positive probability. For every $k_n = o(n)$, its first $k_n$ ranked outdegrees and the first $k_n$ ranks of $n$ independent $X$ observations have total-variation distance tending to zero.*

*Proof.* The number of zero entries in a degree word is invariant under cyclic shifts, just as its multiset of degrees is. Hence the cyclic proof of Corollary 6.4 remains valid after imposing exactly $e_n$ zeros. The tree's degree multiset has the law of the independent sample conditioned

on $(S_n, E_n) = (n-1, e_n)$. Since $\mathbb{E}X = 1$, these targets lie in a bounded central window. Apply Corollary 9.4. □

The condition on $e_n$ is essential to the formulation: a noncentral leaf proportion generally changes the reference offspring law. Corollary 9.5 is a total-variation statement about ranked degrees, not about the geometry or depth-first locations of the tree. It complements the fixed-outdegree count limits in [14] without identifying them with the present assertion.

## 10. Limits of the conclusions

The next two examples distinguish the growing-rank statement from a comparison of the entire sample, and the fixed-law statement from a mean-matched moving tilt.

**Proposition 10.1** (A positive fraction of ranks can retain the constraint)**.** *For every $\alpha \in (0, 1]$, there is a fixed nonnegative integer law of unbounded support, with an exponential moment, for which the distance in* (1.2) *tends to one when $k_n = \lfloor \alpha n \rfloor$ and $m_n - n\mu = O(1)$. Thus $k_n = o(n)$ is the largest distribution-free order of ranks covered by such a statement.*

*Proof.* Choose $p \in (0, \alpha)$ and let

$$\mathbb{P}(X = 0) = 1 - p, \qquad \mathbb{P}(X = j) = p\, 2^{-j}, \quad j \geq 1.$$

This law has span one, mean $2p$ and positive finite variance. Put $m_n = \lfloor 2pn \rfloor$. If $N_n^+$ is the number of positive entries, then $N_n^+ \sim \mathrm{Bin}(n, p)$ and

$$\mathbb{P}(N_n^+ > k_n) \leq e^{-cn}$$

for some $c > 0$ and all large $n$ (for $\alpha = 1$ the probability is zero). The local limit theorem gives $\mathbb{P}(S_n = m_n) \asymp n^{-1/2}$, so the same event has conditional probability $O(\sqrt{n} e^{-cn})$.

On $N_n^+ \leq k_n$, all positive observations are among the top $k_n$ and $\sum_{j \leq k_n} X_{(j)} = S_n$. Thus the event

$$E_n = \left\{ \sum_{j \leq k_n} X_{(j)} = m_n \right\}$$

has conditional probability tending to one, while its unconditional probability is at most $\mathbb{P}(S_n = m_n) + \mathbb{P}(N_n^+ > k_n) \to 0$. Testing total variation on $E_n$ proves the assertion. □

**Proposition 10.2** (The fixed-tilt logarithmic scale)**.** *The assumption $m_n/n \to \lambda$ alone does not imply fixed-tilt approximation even for the largest allocation. In general, the $o(1/\log n)$ centering in Theorem 7.2 cannot be replaced by $O(1/\log n)$.*

*Proof.* Take $w_j = 1$ for all $j$. At $\tau = 1/2$, the tilted law is $\mathbb{P}(X = j) = 2^{-j-1}$, of mean one and variance two. Fix $c > 0$ and put

$$m_n = n + \left\lfloor \frac{cn}{\log n} \right\rfloor.$$

The mean-matched parameter is $\tau_n = m_n/(m_n + n)$, and

$$\log(2\tau_n) = \frac{c}{2 \log n} + o(1/\log n).$$

Along $n = 2^h$, the independent maximum under $p_{1/2}$ satisfies

$$\mathbb{P}(X_{(1)} < h) = (1 - 2^{-h})^{2^h} \longrightarrow e^{-1}.$$

Under $p_{\tau_n}$, $n\mathbb{P}(X \geq h) = 2^h \tau_n^h \to \exp\{c/(2 \log 2)\}$. Its maximum is therefore below $h$ with probability tending to $\exp\{-\exp(c/(2 \log 2))\}$, different from $e^{-1}$. Theorem 6.2(ii) transfers this second limit to the allocation maximum. The difference gives a positive lower bound for the fixed-tilt total-variation distance along this subsequence. □

The assumptions excluding sparse and infinite-variance limits are substantive. The theorems do not treat a degenerate limiting row law, supercritical condensation without a suitable finite-variance tilt, or sums in a non-Gaussian stable domain. The paper does not answer the adjacent sparse-allocation question, Problem 19.9 of [9].

## Appendix A. Finite identities and reproducible checks

The proofs are analytic. A companion program, `code/verify.py`, checks their finite identities in rational arithmetic. It enumerates product samples, groups them both by their indexed rare records and by their rare multisets, and compares the directly conditioned probabilities with (4.4). It also verifies the density normalization, the one-sided total-variation identity (4.5), tilt invariance of conditioned laws, lattice compatibility, and the cyclic-shift identity used for tree degrees. A separate exact finite calculation tests the need to retain the event that the maximum exceeds a truncation level.

The tests include a translated lattice, a law with negative values, a non-upper-tail rare set, an empty rare set, and all attainable totals in each finite model. The threshold-padding test includes a large boundary atom, support gaps and an upper-overflow cutoff. Table 1 records the actual enumeration sizes; every tested identity holds exactly.

Table 1. Exact finite checks in the accompanying program.

| Checked object | Number |
|---|---|
| Product configurations in five record tests | 7,411 |
| Conditional density equalities | 38,871 |
| Conditional tilt-invariance equalities | 1,024 |
| Degree words for $2 \le n \le 8$ | 4,706 |
| Residual-centering identities | 891 |
| Threshold-padding identities | 48,710 |

These computations test finite algebra and representations; they do not prove a limiting theorem or a convergence rate. The program uses only the Python standard library and writes its results to `code/checks.json`. It is run by

```
python code/verify.py -output code/checks.json.
```

The separate program `code/verify_extensions.py` checks 23,102 vector conditional-density identities at 192 attainable conditioning totals, 5,632 centered residual-sum identities, and 35,888 cyclic shifts preserving the leaf count. It also checks the exact equality between the total-variation distances of full rare-value counts and their total counts in a finite categorical model. The output is recorded in `code/extension_checks.json`.

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This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [15] and working together inside the Brain framework [17], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [16], the Math Brain repository [15] and the Brain framework repository [17].

## Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

## Declaration of generative AI and AI-assisted technologies

AI systems produced the original mathematical arguments, calculations, verification programs, and text. The author initiated and organized the research programme and arranged its communication; the author did not supply mathematical arguments or calculations and has not conducted a human mathematical review of this work.

The mathematical development began with a rare-cloud argument for finitely many ranks and a direct-conditioning argument. ChatGPT (OpenAI) developed the indexed-record formulation, the uniform local-limit and censoring proofs, and the extensions to Poisson counts, growing coordinate blocks, and several lattice constraints; it also carried out literature comparisons, proof checks, and exact finite computations. These checks are not independent human peer review or an end-to-end formal verification.

This work forms part of *Math Brain*, a research programme on the ability of AI systems to solve mathematical problems. Details of the project will be published separately. The author is responsible for the manuscript and decisions concerning its dissemination.

## Data and code availability

The manuscript, LaTeX source, verification programs, recorded outputs, and a literature-comparison note are available at

`https://github.com/yspennstate/joint-allocation-extremes`.

No empirical dataset is used. The verification programs use only the Python standard library. The mathematical results are proved analytically and do not depend on the computations.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [15].

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Einstein Institute of Mathematics, The Hebrew University of Jerusalem, Jerusalem, Israel
*Email address*: Yitzchak.Shmalo@mail.huji.ac.il